

CIS 635: Assignment 1

Exploratory Data Analysis of Climate Change Data

Deadline: Please check the Blackboard post

Total points: 10

Objective

This assignment provides hands-on practice with **data preparation, descriptive statistics, exploratory data analysis (EDA), and data visualization**. You will work with climate-change data and use numerical summaries and visualizations together to identify patterns, variability, and relationships in the data. The data are samples; do not make global conclusions from these samples.

Data

Use the following two files:

- **Climate distortion data:** https://raw.githubusercontent.com/mdkamrulhasan/data_mining_kdd/refs/heads/main/data/climate-change/climate_distortion.csv
- **Country-region data:** https://raw.githubusercontent.com/mdkamrulhasan/data_mining_kdd/refs/heads/main/data/climate-change/country_region.csv

Tasks

(a) Data Preparation and Initial EDA — 1.5 points

- Identify the number of columns containing NULL values in either data file and briefly describe the pattern of missingness.
- Impute the NULL values using a reasonable method/logic and briefly justify your choice.
- Report the number of unique Region(s) recorded for the USA.

(b) Descriptive Statistics — 2.5 points

For the USA data, analyze C02 Emissions MT and Pesticide Use KG per HA over the available period.

- Report the mean, median, variance, standard deviation, minimum, and maximum.
- Report Q_1 , Q_3 , and the interquartile range (IQR), where

$$\text{IQR} = Q_3 - Q_1.$$

- Briefly compare the two variables using the numerical summaries. What do the statistics suggest about their center and variability?

(c) Exploratory Data Analysis and Relationships — 3 points

Use the data to investigate the following questions:

- Which country had the most and the least amount of C02 Emissions MT before and after 2000? Report the corresponding values. (1)
- What was the coolest date recorded in the USA? (0.5)
- Which Crop Type from the “Midwest(U)” and “Northeast(U)” regions together had the highest historical Economic Impact Million USD? (1)
- For the eight variables below, identify one pair with the highest association and one pair with the lowest association. If there is a tie, choose one pair. Briefly explain what the association suggests. (0.5)

Average Temperature C, Total Precipitation mm, CO2 Emissions MT, Extreme Weather Events, Pesticide Use KG per HA, Fertilizer Use KG per HA, Soil Health Index, and Economic Impact Million USD.

(d) Visualization and Interpretation — 3 points

Create clear, properly labeled visualizations that support your EDA.

- (i) Create a histogram for each of CO2 Emissions MT and Pesticide Use KG per HA for the USA. Use the plots to describe the distribution shape (e.g., approximately symmetric, skewed, bimodal, or heavy-tailed). (1)
- (ii) Create a graph showing the yearly average Total Precipitation mm in the USA and briefly describe any visible trend or pattern. (1)
- (iii) Create a scatter plot for one variable pair from part (c)(iv). Use the visualization to support your interpretation of the association. (1)

Submission Guidelines

Submit a runnable notebook (Python, R, or another programming language of your choice) through Blackboard (**the link of your notebook, which stays in your Google drive or else where, is not accepted**). Include sufficient markdown/comments to explain your analysis and decisions. Use appropriate labels, titles, units, and legends in visualizations. Report numerical results to three decimal places where appropriate.

Important: The goal is not full test-style coverage of every possible data operation. Focus on meaningful analysis: use **descriptive statistics, EDA, and visualization together** to support your observations and conclusions.